

TOPIC 3: COMPONENTS OF A COMPUTER NETWORK

1. Key Devices

- **Server** – a powerful computer that provides services (files, websites, email) to other computers.
- **Client** – a computer or device that requests services from a server (e.g. your laptop).
- **Switch** – connects devices within a LAN and forwards data to the correct device.
- **Router** – connects different networks (e.g. your home LAN to the internet) and directs data between them.
- **Hub** – a simpler (older) device that connects devices in a LAN but broadcasts data to all ports.

2. Network Media

- **Cables** – e.g. Ethernet (twisted pair), fibre optic; reliable and often faster for fixed connections.
- **Wireless** – Wi-Fi, radio; convenient for mobile devices; signal can be affected by distance and obstacles.

3. Summary

Networks need servers, clients, switches/routers and media (cables or wireless) to carry data. Understanding these components helps you see how school and home networks work.

End of Topic 3